

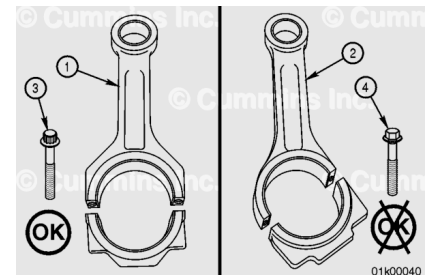
Trainings ▾

Remove

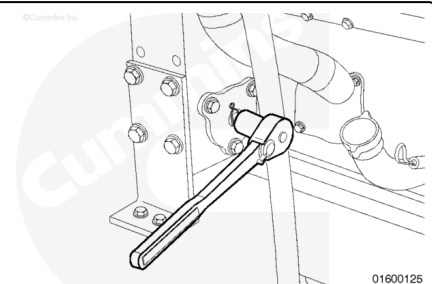
The following steps **must** be used for engines containing the straight split connecting rod **only** (1).

Use the following procedure for angle split connecting rods (2). Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

The straight split connecting rod can be identified by 12 point head capscrews (3).

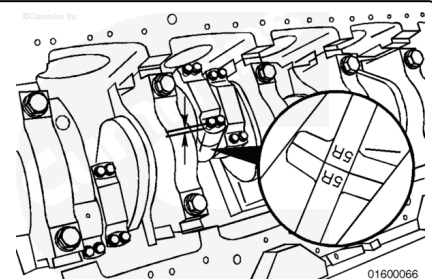


Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.



The connecting rods **must** have the cylinder number and bank marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for the correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

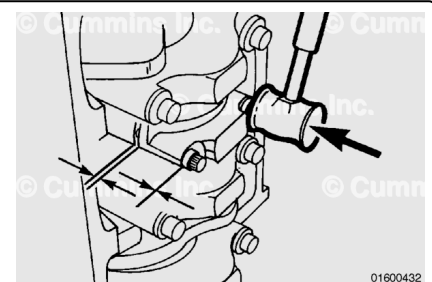
Loosen the capscrews until there is 6 mm [0.236 in] of clearance between the rod cap and the capscrew head.



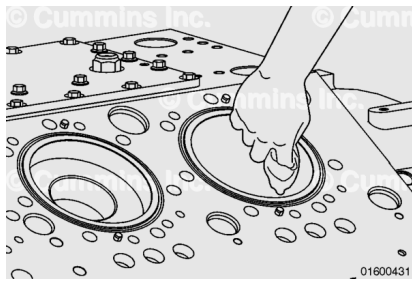
Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate.

Tighten the capscrews by hand.

This step is used to break the connection between the connecting rod and cap. **Do not** remove the cap at this time.

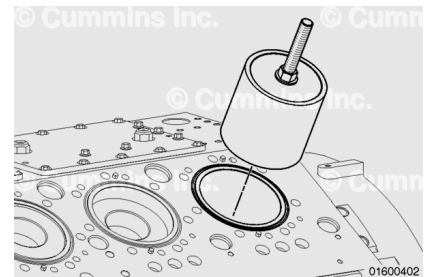


Remove any excess residual oil from the liner bore.



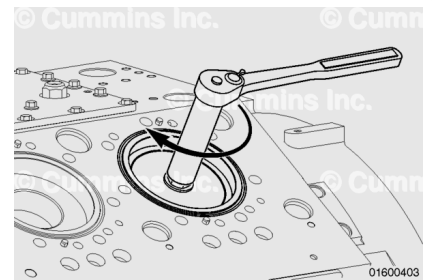
Insert the plunger into the cylinder liner until it contacts the piston.

Once inserted, the top face of the plunger will be approximately 20 mm [0.79 in] below the top of the liner.

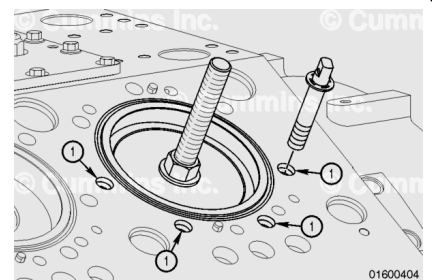


Tighten the plunger to lock the plunger into the liner.

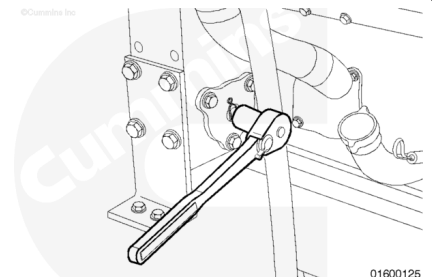
Torque Value: 125 n·m [92 ft-lb]



Insert the four block guide studs into the block at the locations shown (1) until the circlip contacts the block face.



Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at top dead center.



As the connecting rod moves up to top dead center, the power cylinder assembly is pushed out of the block.

If the interference fit between the liner and the block is excessive, the plunger may be pushed up without removing the liner. If this occurs, reference the "Hydraulic Power Cylinder" section below for power cylinder removal.



The tooling described in the following section of the procedure is for lifting **only**. It should **not** be used to break the cylinder block adhesion to the upper press fit area and packing bore seals.

Install the four frame tube legs and lock them to the guide studs with the attached pins.

Note : In applications with sufficient overhead clearance, the additional extension frame tube legs can also be fitted.

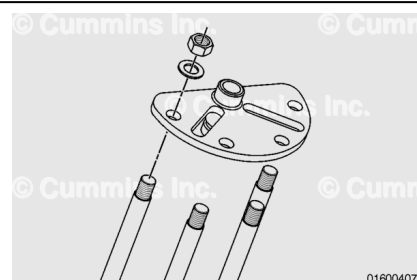


⚠ CAUTION ⚠

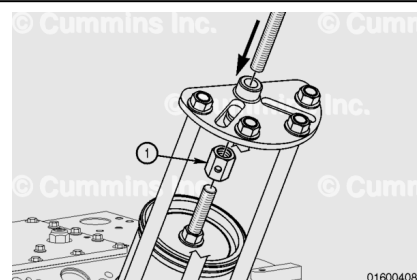
Do not over-tighten the nuts or tool damage can occur.

Attach the top plate with the boss facing upwards.

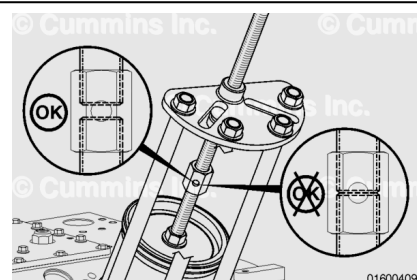
Secure the plate with the four nuts and washers provided.



Insert the threaded bar through the top plate boss and attach it to the plunger using the hexagonal coupling (1).

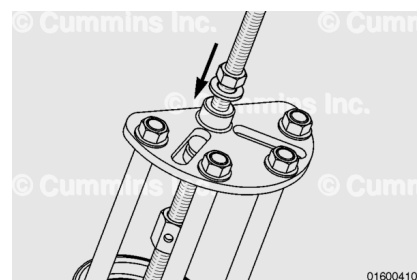


There should be clearance through the locking pin hole in the hexagonal coupling.



Install the thrust washer and extraction nut to the threaded bar. Hand-tighten against the top plate.

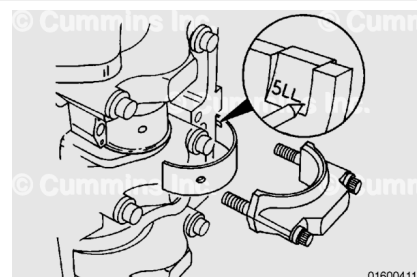
The power cylinder is now locked in place.



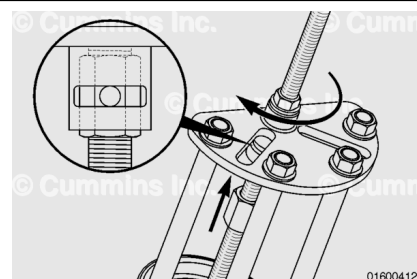
Remove the connecting rod capscrews and the cap.

Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

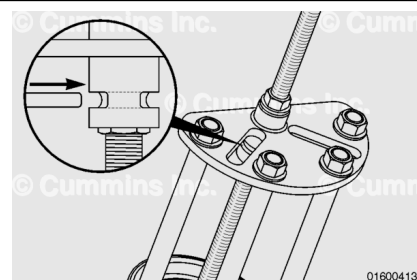
Mark the bearing for future identification or analysis.



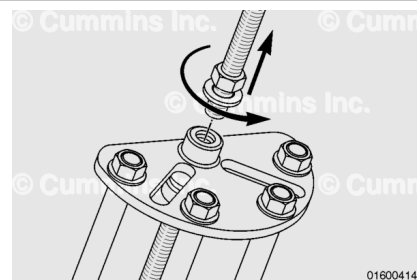
Turn the top nut **clockwise** to draw the power cylinder from the block until the hole in the hexagonal coupling aligns with the hole on the underside of the top plate.



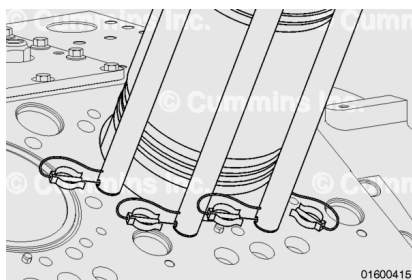
Insert the locking pin through the plate and hexagonal coupling.



Loosen the top nut and remove the threaded bar from the hexagonal coupling and top plate.

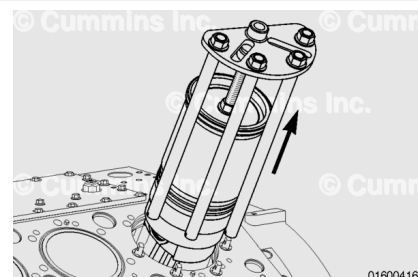


Remove the four locking pins from the lower guide studs.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



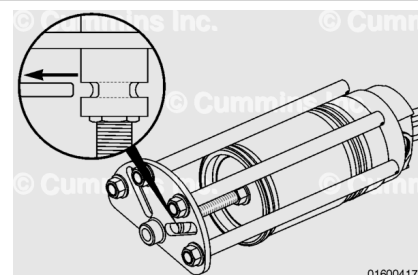
⚠ CAUTION ⚠

Piston ring or liner damage could relieve the vacuum holding the piston in place causing it to fall from the assembly resulting in personal injury or component damage.

Remove the power cylinder assembly.

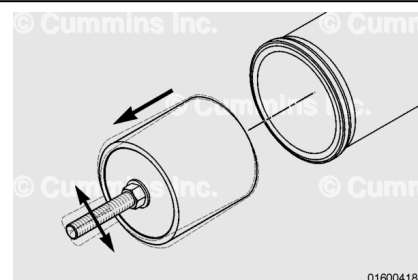
Place the assembly on a pallet or bench.

Remove the top locking pin from the hexagonal coupling to release the lifting assembly.



Loosen the plunger nut and gently pull to release it from the liner bore. A side-to-side motion may be required to break the seal between the liner and plunger.

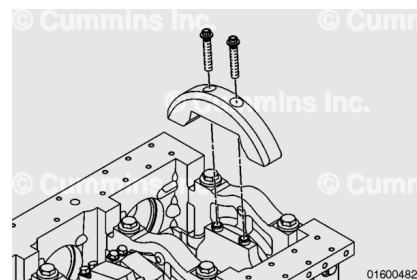
Remove the plunger from the liner.



Hydraulic Power Cylinder

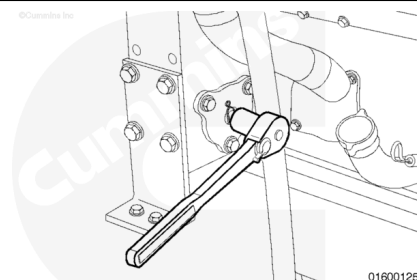
Working from the bottom of the engine, remove the counterweight, if required. Refer to Procedure 001-016 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-016-

tr.html)

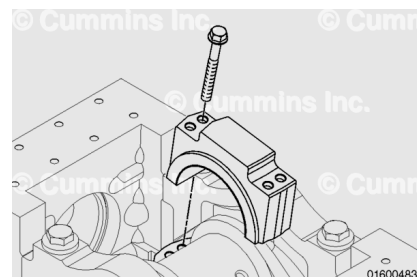


Use the barring mechanism to rotate the engine.

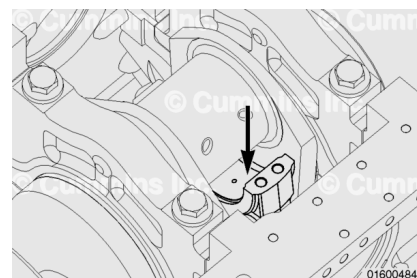
Rotate the crankshaft to position the connecting rod at bottom dead center.



Remove the connecting rod capscrews and the rod cap.

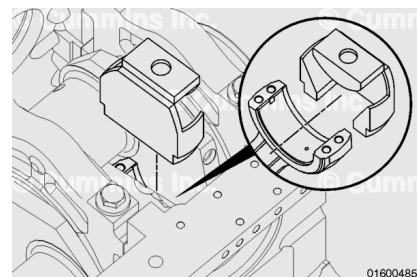


Push the piston and connecting rod away from the crankshaft.

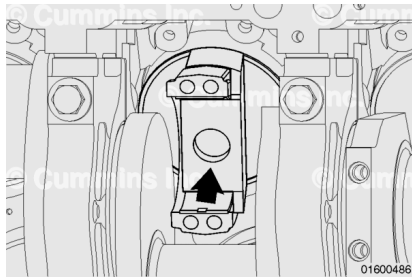


Attach a cylinder liner removal tool to the connecting rod.

Note : The crank may need to be rotated slightly to improve access.



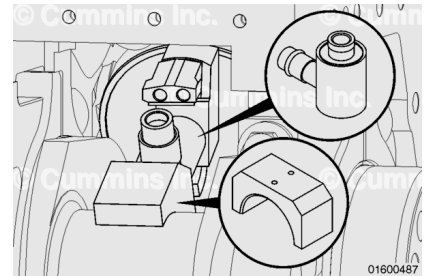
Push the connecting rod and piston all the way up the piston liner until the recess of the cylinder liner removal tool engages the liner.



If rotated earlier, rotate the crankshaft back to bottom dead center.

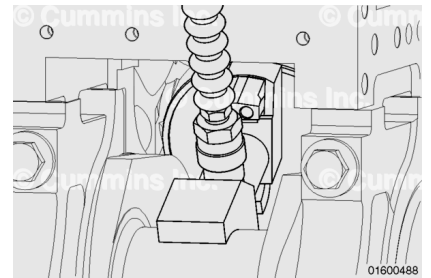
Install the locating cup and hydraulic cylinder onto the crankshaft connecting rod journal.

Note : The locating cup and hydraulic cylinder are bolted together as one piece.

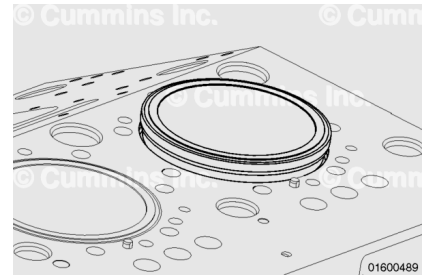


Attach a hydraulic line to the hydraulic cylinder.

Gently apply pressure until the hydraulic ram engages with the cylinder liner removal tool.



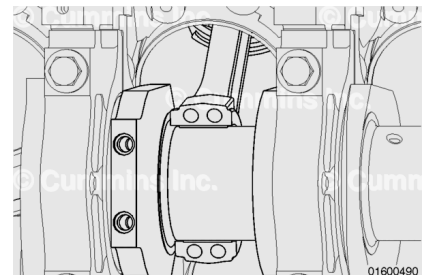
Press the liner out until the liner interference fit is cleared and the liner is loose.



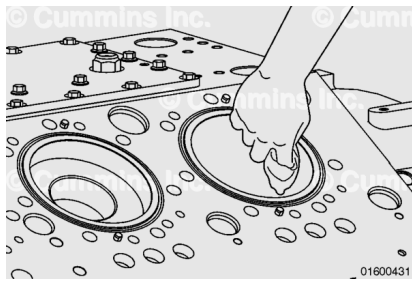
Remove the hydraulic cylinder, remover plate, and cylinder liner removal tool from the engine.

With these removed, push the piston back down the liner and guide the connecting rod back onto the crankshaft. The crankshaft should be at bottom dead center.

Note : The connecting rod **must** be pushed down to clear the liner for removal of the cylinder liner removal tool.



Remove any excess residual oil from the liner bore.



Insert the plunger into the cylinder liner until it contacts the piston.



Tighten the plunger to lock the plunger into the liner.

Torque Value: 125 n•m [92 ft-lb]

Continue to remove the power cylinder as described above.

